

AKSHAT SAINI

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EDUCATION

B.Tech, Computer Science Engineering (AI/ML Specialization)
Bennett University

Apr 2027
CGPA: 7.37

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React.js, Next.js, Node.js, Express.js, TensorFlow, Keras, Scikit-learn, OpenCV

Databases & Tools: Neo4j, GitHub, MongoDB, MySQL, DuckDB, Git, GCP, Vercel, Pandas, NumPy

Domains: AI/ML, RAG Systems, Backend/API Design, Full-Stack Development

EXPERIENCE

Machine Learning Intern
Fox Trading (Remote)

Sept 2025 – Nov 2025

- Designed and deployed end-to-end **ML pipelines** for Diabetes Prediction, Human Action Detection, and Bike Demand Forecasting, including **data preprocessing**, **feature engineering**, and model evaluation.
- Implemented ML models using **Scikit-learn**, **TensorFlow**, **Keras**, and **MediaPipe**, improving prediction.

PROJECTS

Promcom – Graph-Based Dating & Networking Platform (JWT, Microsoft OAuth, REST APIs)

[GitHub](#)

- Architected a graph-driven matchmaking platform using **Neo4j graph database**, modeling node-relationship structures and writing optimized **Cypher queries** to compute **compatibility matching** via shared interests, mutual connections, and behavioral attributes.
- Built a full-stack application with **Next.js**, **Node.js**, and **Express**, implementing secure **authentication & authorization** via **JWT** and **Microsoft OAuth** for identity management and session handling.
- Engineered a **recommendation system** with efficient **graph traversal**, supporting 20+ concurrent users with minimized query latency; integrated **Cloudinary** for media handling and deployed on **Vercel**.

GeoSence–Geopolitical Intelligence Platform (Streamlit, Vector Embeddings, Semantic Search)

[GitHub](#)

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to process the **GDEL T 1.0 dataset (240GB)**, building **ETL workflows** to transform raw event streams into columnar **Parquet** format for high-performance querying via **DuckDB** and **ClickHouse Cloud**.with dynamic **k-NN Retrieval**
- Implemented **vector similarity search** using the **all-MiniLM-L6-v2** embedding model for semantic retrieval over large-scale geopolitical event data, enabling context-aware **knowledge base construction**.
- Integrated **Google Gemini API** with **prompt engineering** for **contextual generation**, synthesizing retrieved events into structured geopolitical analysis; containerized with **Docker** for reproducible, scalable local deployment.

Hand Sign Detection–Sign Language Interpreter (Python, TensorFlow, Transfer Learning, NumPy)

[GitHub](#)

- Built a **computer vision** system using **transfer learning** to fine-tune a **MobileNetV2** architecture for **image classification** of static hand gestures (A–Z, 0–10) from the **Kaggle American Sign Language Dataset**.
- Developed a real-time inference pipeline with **OpenCV** for frame capture/preprocessing and **TensorFlow**-based **gesture recognition**, achieving **86% accuracy** through **data augmentation**, dataset normalization, and hyperparameter tuning.
- Optimized **model fine-tuning** and **live camera inference** for low-latency, real-time gesture-to-text translation suitable for live interpretation scenarios.

POSITION OF RESPONSIBILITY

Co-Lead, Google Developer Group (GDG)

Jan 2023 – Jul 2026

- Led **70+ member team**; organized **9+ events** engaging **4,000+ students**; managed speakers and sponsorships.

Management Lead, Bennett Cloud Computing Club

Aug 2024 – Aug 2025

- Directed PR and operations for cloud-focused technical events.

ACHIEVEMENTS

- Solved **200+ LeetCode problems** across **Data Structures & Algorithms**, covering **Graphs**, **Dynamic Programming**, **Trees**, **Greedy**, and Advanced Data Structures.
- Secured **1st Place** in the **Web3 Track** at **Hackaccino 4.0 (2026)** for building an innovative blockchain-powered solution.